PORTFOLIO PROJECT

Big Data Architecture & Analytics

HEALTHCARE INSURANCE WITH AI

An End-to-End Big Data Architecture, Normalization Strategy, & GenAI Semantic Layer

TOOLS & TECHNOLOGIES USED

- **Python** — Pandas & Chunking
- **MySQL** — Workbench & Schema
- **Claude 3.5** — GenAI Semantics
- **Power BI** — Desktop & DAX

PROJECT SCOPE

Introduction: What Are We Analyzing?

Core Objective

Mapping multi-state health insurance premium variations across complex **US Marketplace infrastructure**.

The Business Problem

Payer records are traditionally **multi-gigabyte, un-normalized**, and dense with technical jargon.

Analytical Shift

Converting structural text into **plain-language insights** to streamline stakeholder decision-making.

High-Impact Industry Utility: Healthcare Consulting | Pharma Market Intelligence | Software Vendors

MARKET INSIGHTS

Key Findings & Domain Intelligence

Availability Index

High-cost specialized therapeutics and biological metrics are **heavily restricted** and clustered within top-tier state networks.

Premium Multipliers

Insurance plans offering comprehensive genetic screening options charge **significantly higher baseline** standard rates.

Tobacco Penalty Variations

Extensive geographic gaps reveal multiple states where carriers completely **bypass extra tobacco billing penalties**.

Dynamic Cross-Filtering

Financial variables dynamically recalculate across visualization cards as users interact with **state geographical map filters**.

PIPELINE ENGINEERING

Technical Architecture & Data Pipeline



DATA ENGINEERING

Relational Key Mapping & Optimization

THE DISCREPANCY

Empty Table Joins

Standard **INNER JOIN** queries failed due to structural string length differences across datasets.

Benefits: Variant ID (16 Chars)
Rates: Base ID (14 Chars)

THE SOLUTION

Trimmed Join Predicates

Utilized SQL predicates to dynamically align millions of cost metrics with clinical benefits.

```
LEFT(TRIM(PlanId), 14)
```

AI LAYER

Claude 3.5 Automated Semantics

1

The Cost Challenge

Executing live API calls across millions of redundant records is **prohibitively expensive** for enterprise analytics pipelines.

2

Asymmetric Caching

Isolated unique benefit descriptions, run AI summarization once, and cached results into a fast lookup relational layer.

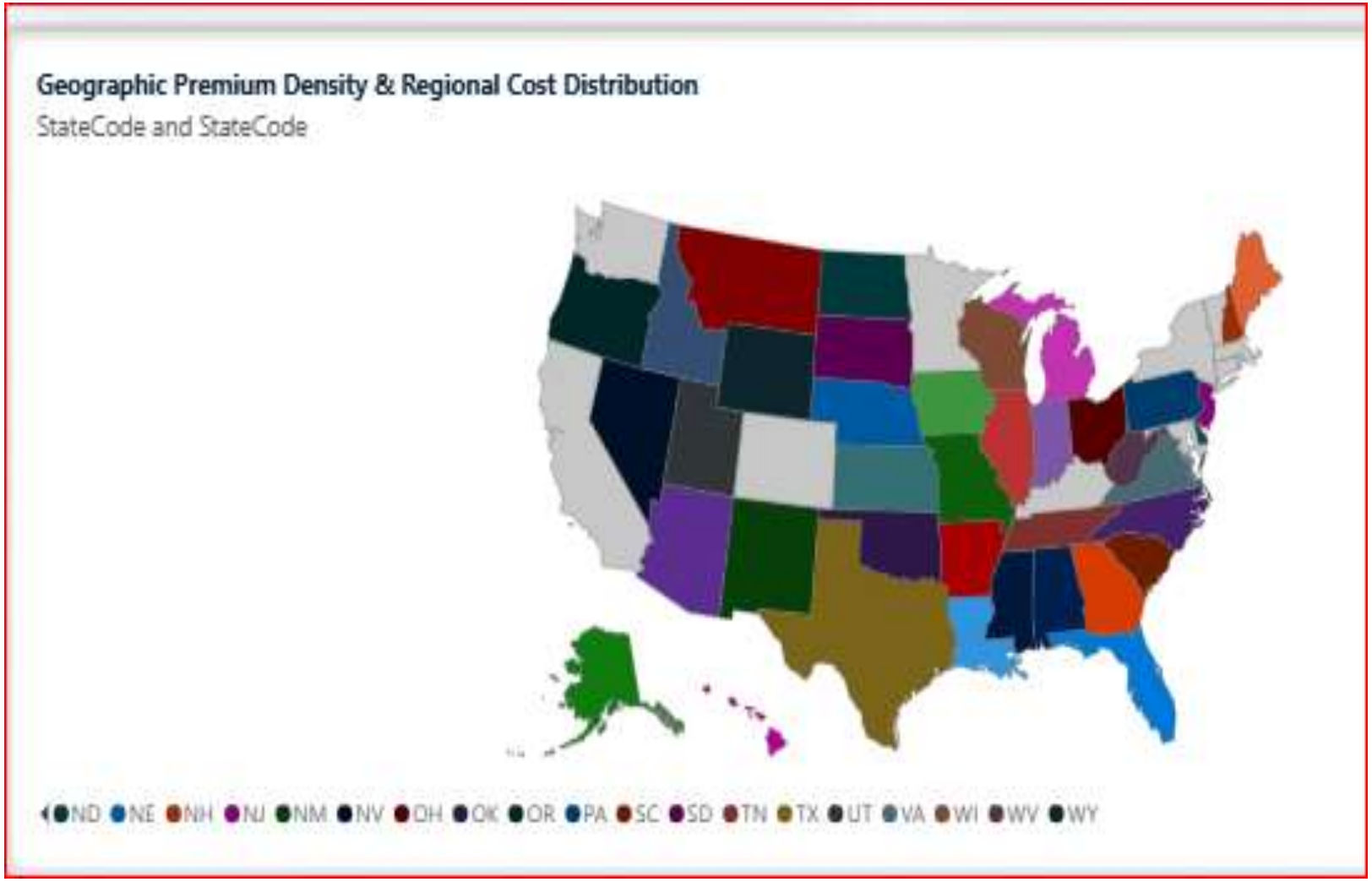
3

DAX Tooltip Safeguards

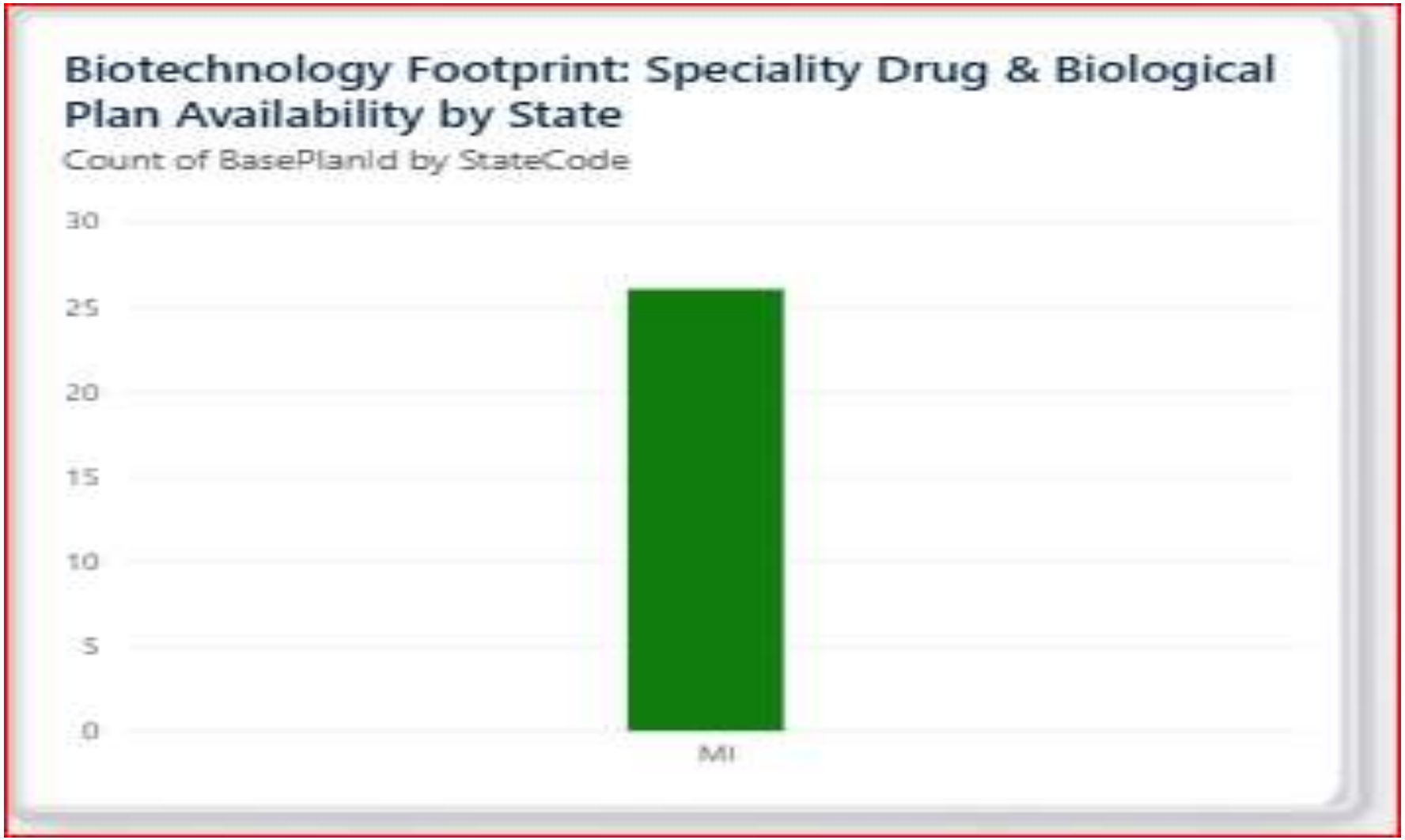
Engineered logic to gracefully present unified branded tooltips without dashboard breaks.

```
IF(ISBLANK())
```

Power BI Report & Visual



Power BI Report & Visual



CONCLUSION

Project Conclusion & Business Summary

Data Validation

Proved messy multi-gigabyte databases can be optimized into **responsive cloud BI models**.

AI Business Value

Translates legal insurance jargon into **transparent patient-first actionable data**.

Production Readiness

Ready to switch from local mock data layers straight into **paid active API endpoints**.



THANK YOU

Full-Stack Portfolio Project Completed Successfully

Let's connect to optimize your healthcare data systems!